

heat and Fokker-Planck equations can
be recovered as the gradient flows of certain energies



Unified framework for many PDEs







THE VARIATIONAL FORMULATION OF THE FOKKER–PLANCK EQUATION*

RICHARD JORDAN[†], DAVID KINDERLEHRER[‡], AND FELIX OTTO[§]

In memory of Richard Duffin

Abstract. The Fokker–Planck equation, or forward Kolmogorov equation, describes the evolution of the probability density for a stochastic process associated with an Ito stochastic differential equation. It pertains to a wide variety of time-dependent systems in which randomness plays a role. In this paper, we are concerned with Fokker–Planck equations for which the drift term is given by the gradient of a potential. For a broad class of potentials, we construct a time discrete, iterative variational scheme whose solutions converge to the solution of the Fokker–Planck equation. The major novelty of this iterative scheme is that the time-step is governed by the Wasserstein metric on probability measures. This formulation enables us to reveal an appealing, and previously unexplored, relationship between the Fokker–Planck equation and the associated free energy functional. Namely, we demonstrate that the dynamics may be regarded as a gradient flux, or a steepest descent, for the free energy with respect to the Wasserstein metric.

Key words. Fokker–Planck equation, steepest descent, free energy, Wasserstein metric

Historical Context & Importance of JKO

- **Jordan–Kinderlehrer–Otto (1998):**
 - Introduced the variational formulation of diffusion as a gradient flow in Wasserstein space.
 - Showed that the **heat and Fokker-Planck equations can be recovered as the gradient flows of certain energies.**
- **Impact:**
 - Sparked development of “Otto calculus”: treating probability measures as an (∞ -dim) Riemannian manifold.
 - Provided a powerful bridge between PDE theory, optimal transport, and variational methods.
- **Practical Importance:**
 - **Unified framework for many PDEs** (heat, Fokker–Planck)
 - Basis for many modern numerical schemes (proximal / variational time-stepping).
 - Influential in ML & generative modeling: diffusion models, regularized flows, entropy minimization.



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Gradient Flows: Big Picture

- Many PDEs = gradient flows of energies $\mathcal{E}(\rho)$
- In Wasserstein space $(\mathcal{P}(\mathbb{R}^d), W_2)$, the JKO scheme makes this precise.
- Key idea: PDE evolution = iterated minimization of energy + transport cost.